



Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch : V Semester B.E. ECE A
Course Code & Title : EC3501 Wireless Communication
Name of the Faculty member : Mr.G.Sivakumar

Innovative Practice Description

- **Unit / Topic:** Unit V/ Development of Wireless Networks
- **Course Outcome:** CO5
- **Unit Outcome:** 5a
- 1 **Activity Chosen:** Reciprocal Peer Questioning
- **Justification:**

It provides students with higher order open-ended questions to generate a focused discussion in a small group setting. The students will be able to understand the evaluation of wireless networks with the help of Reciprocal Peer Questioning activity more easily.

- **Time Allotted for the Activity:** 40 Minutes

- **Details of the Implementation:**

Reciprocal Peer Questioning is an activity in which students question each other. First, students were taught with the evaluation of networks. Then, Students are then given a few minutes to individually prepare several content-specific questions.

The questions are generic prompts students use to generate specific content-based questions. After that, the students were encouraged to ask each other questions in order to learn the concepts more easily and quickly.

- **CO – PO / PSO mapping:**

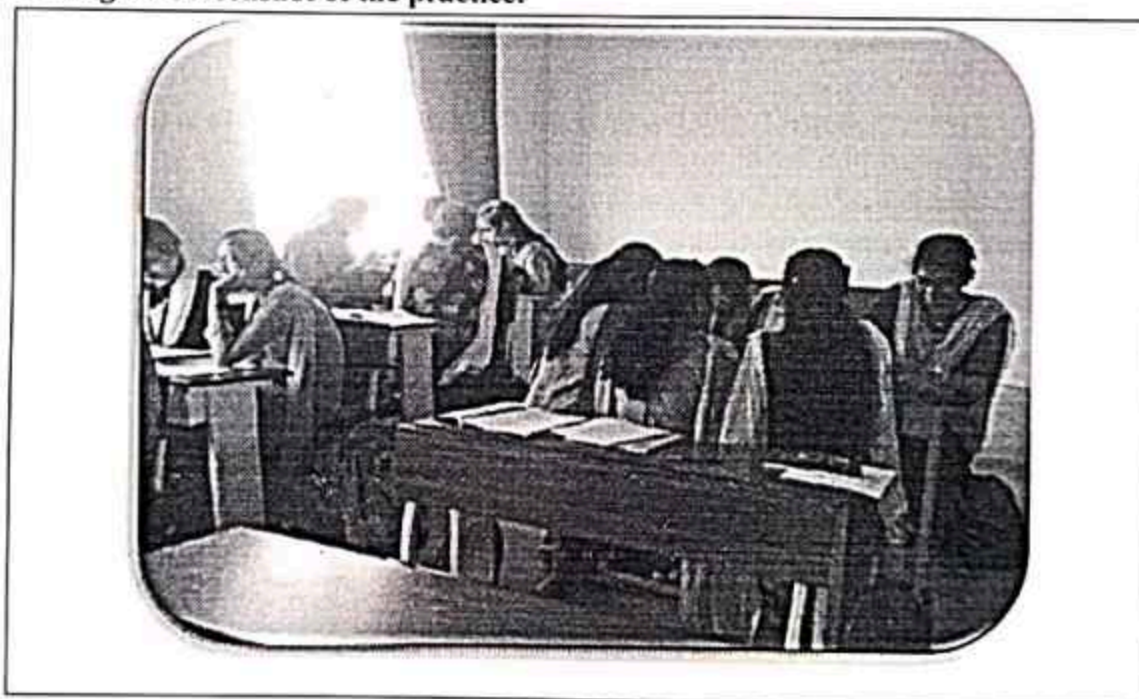
CO	PO 1	PO 2	PO 3	PO 5	PO 9	PO 12	PSO 1	PSO 3
CO3	3	2	2	3	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO 9	PO 12	PSO 3
Justification for correlation	Students will function effectively as an individual, and as a member in Reciprocal Peer Questioning activity to discuss about various Instruction Sets, So Course outcome is mapped at level 2	In Reciprocal Peer Questioning activity students will gain an understanding of the different multiple access techniques used in wireless networks hence Course outcome is mapped at level 2	The students understanding the concepts and structure of wireless networks. So Course outcome is mapped at level 2

• Images / Screenshot of the practice:



• Reflective Critique:

- ❖ *Feedback of practice from students and other stakeholders:*
- ❖ The effectiveness of the activity was felt when the students were asked the questions on the topic, they were able to answer easily. They also understood the concepts more easily and able to remember while writing the test.
- ❖ *Benefit of the practice:*
- ❖ They learnt the concepts and asked more doubts in the topic and learnt many things. Outcome attainment has also been increased since the students' performance in the test is also good.

❖ *Challenges faced in implementation:*

- Each student has to read the material posted in the course website.
- Non participation in the activity: All the students should be made involve in the activity

References:

- ❖ Rashwan, Z. I., Busebaia, T. J., AL-Sabbagh, A. S., & Eweida, R. S. (2021). Effect of guided reciprocal peer questioning strategy on pediatric nursing students' self-esteem and metacognitive awareness: Current approach and future irections. *Nurse Education Today*, 107, 105153. <https://doi.org/10.1016/j.nedt.2021.105153>.
- ❖ Rappaport, T.S., -Wireless communications", Pearson Education, Second Edition, 2010.

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Degree, Semester & Branch : V Semester B.E. ECE A

Course Code & Title : EC3501 Wireless Communication

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Innovative Practice Description

- **Unit / Topic:** Unit IV / Time Division Multiple Access (TDMA)
- **Course Outcome:** CO 4
- **Unit Outcome : 4a**
- **Activity Chosen:** Flipped Classroom
- **Justification:** This allows students to learn at their own pace, encourages them to actively engage with lecture material, frees up actual class time for more effective, creative, and active learning activities, and allows them to be in control of their learning.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 - The topic was given to the students learn on their own. Resources like reference book, PPT and videos were already post in canvas. This flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related
 - In this blended learning approach, face-to-face interaction is mixed with independent study—usually via technology. In a common Flipped Classroom scenario, students might watch pre-recorded videos at home, then come to school to do the homework armed with questions and at least some background knowledge.
 - The concept behind the flipped classroom is to rethink when students have access to the resources they need most. If the problem is that students need help doing the work rather than being introduced to the new thinking behind the work, then the solution the flipped classroom takes is to reverse that pattern.

• **CO – PO / PSO mapping:**

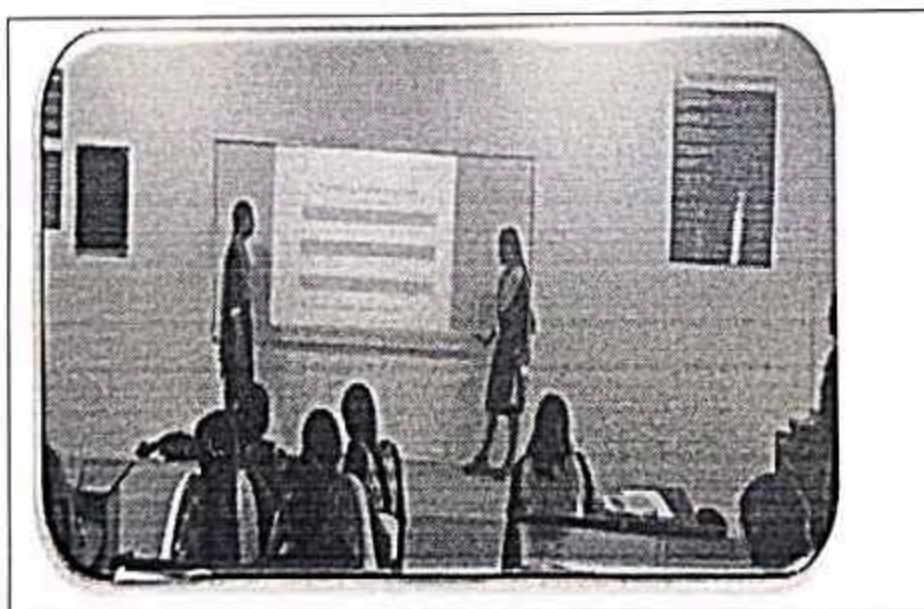
CO	PO 1	PO 2	PO 3	PO5	PO 9	PO 10	PO 12	PSO1	PSO3
CO4	3	2	2	3	2	2	2	1	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO 9	PO 10	PSO3
Justification for correlation	Students will function effectively as an individual, and as a member or leader in Flipped Classroom activity to discuss about different Task Assignment Algorithms, So Course outcome is mapped at level 2	In Flipped Classroom activity students will make effective presentations about TDMA , hence Course outcome is mapped at level 2	Knowledge of TDMA is used to design a wireless Communication system. So Course outcome is mapped at level 2

• Images / Screenshot of the practice:



• Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- Students eagerly participated and enjoyed this activity.

❖ *Benefit of the practice:*

- Outcome attainment would have increased due to innovative practice over conventional practice.
- Students easily understand the concepts of TDMA.
- Students can answer easily about the need of multiple Access Techniques.

❖ *Challenges faced in implementation:*

- Faced issues while make the students to understand about Flipped Class Room Activity rules.
- Make the students to learn about Multiple Access Technique before the start of this activity.

References:

- ❖ Bergmann, J., & Sams, A. (2012). Flip your classroom: Reach every student in every class every day. Eugene, Or: International Society for Technology in Education.
- ❖ Center for Teaching Innovation at Cornell University. (2017). Flipping the classroom. Retrieved from <https://www.cte.cornell.edu/teaching-ideas/designing-your-course/flipping-the-classroom.html>.

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Department of Electronics and Communication Engineering
Academic Year: 2023-2024 (Odd Semester)

Innovative Practices Description

Degree, Semester & Branch: V Semester B.E. ECE A

Course Code & Title: EC3501 Wireless Communication

Name of the Faculty member: Mr.G.Sivakumar

UNIT I		Date: 28.07.2023
Name of the Topic	Name of the Innovative Practice	
Prioritizing Handoffs, Practical Handoff Considerations	Think Pair Share	
		
UNIT II		Date: 29.08.2023
Name of the Topic	Name of the Innovative Practice	
Fading Effects Due To Doppler Spread	Brain Storming	
		

4.5.27.2023
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UNIT III		Date: 20.09.2023
Name of the Topic	Name of the Innovative Practice	
Equalization, Diversity and Channel Coding- Introduction, Fundamentals of Equalization	Zero Minute Speech	
		

UNIT IV		Date :16.10.2023
Name of the Topic	Name of the Innovative Practice	
Time Division Multiple Access (TDMA)	Flipped Class Room	
		


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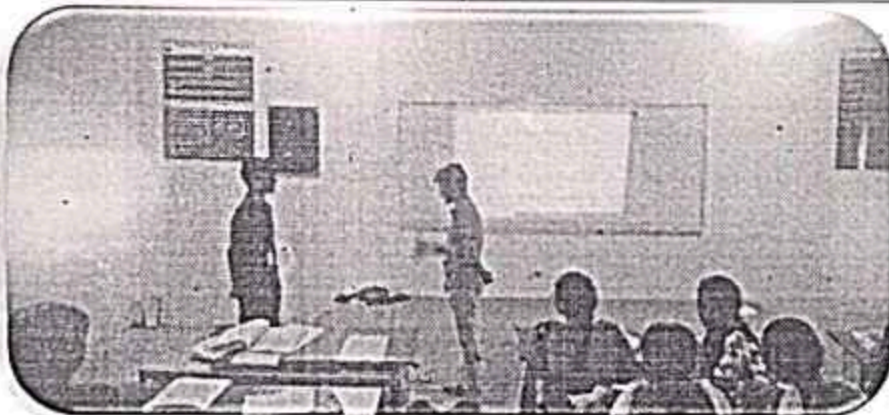
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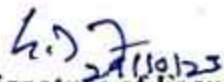
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UNIT V		Date :29.10.2023
Name of the Topic	Name of the Innovative Practice	
Development of Wireless Networks- First Generation Wireless Networks, Second Generation Wireless Networks, Third Generation Wireless Networks	Reciprocal Peer Questioning	
		


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